IEEE CS NEXTGEN: JUNIORS AT CONFERENCE 2026 · AI & ROBOTICS

Remember What Got Fixed

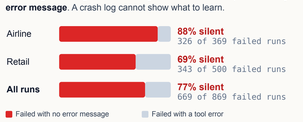
Emotion-Shift Memory saves only the past chats where a frustrated customer ended up satisfied. With that 400-token memory, an AI agent repeated 27% fewer mistakes on tasks it had never seen.

REAL HUMAN CHATS	600 (WildFeedback)
AI AGENT RUNS	2,624 (tau2-bench)
MEMORY SIZE	400 tokens of plain text
CODE	1,218 lines of Python

01 · THE PROBLEM

Most AI mistakes make no noise

We studied 2,624 runs of AI agents helping simulated airline and store customers. When a run failed, the agent usually broke a store rule (refunding the wrong card, cancelling a ticket it should not) **without any error message**. A crash log cannot show what to learn.



02 · THE IDEA

Let the customer's mood pick the lesson

Two yes/no questions sort every finished chat into one of four boxes. Only one box holds both a mistake *and* the fix that worked.

	END: SATISFIED	END: STILL FRUSTRATED
MID-CHAT: FRUSTRATED	**Frustrated → Satisfied** Has the mistake and the fix. `Save 1 rule`	**Frustrated → Frustrated** The fix never came. `Skip`
MID-CHAT: CALM	**Calm → Satisfied** Nothing went wrong. `Skip`	**Calm → Frustrated** Broke at the end, no fix. `Skip`

03 · HOW IT WORKS

Five steps, no vector database

1. A chat with a customer ends.
2. A mood detector puts the chat in one of the 4 boxes.
3. For **Frustrated → Satisfied** chats, a rule writer (Gemini) writes one line: When … `AVOID … INSTEAD …`
4. The line is added to `MEMORY.md`, a plain text file.
5. In each new chat, keyword matching copies the 6 best-matching rules (at most 400 tokens) into the agent's instructions.

04 · A REAL HUMAN CHAT (WILDFEEDBACK WF_129750)

Word lists miss polite frustration

PERSON ASKS THE AI

"give me 10 sentences that end in the word apple"

PERSON'S REPLY TO EACH AI ANSWER	HUMAN LABEL	VADER	LAYA
"you got 7 out of 10"	dissatisfied	+0.00	52%
"2 out of 3"	dissatisfied	+0.00	31%
"finally"	satisfied	+0.00	16%
Box chosen	**Frustrated → Satisfied**	**Missed**	**Found**

VADER scores a reply from −1 (negative) to +1 (positive), so 0.00 means it found nothing negative in "you got 7 out of 10." Laya reads each reply together with the AI answer before it and gives the chance the person is frustrated.

77% of failed AI runs showed no error (669 of 869)

59% vs **45%** F1 at spotting "frustrated → satisfied" chats: Laya vs a word list

44% → 32% repeated mistakes: no memory vs Emotion-Shift Memory (27% fewer)

400 tokens of memory per chat, for every condition. No fine-tuning, no vector database.

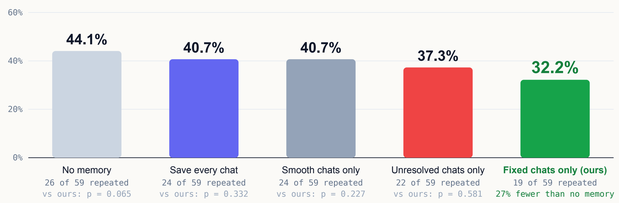
05 · MAIN EXPERIMENT

Which past chats should go into MEMORY.md?

KEPT THE SAME: Agent: Gemini 3.8 Flash · Mood detector: Gemini 3.5 Flash-Lite · Same rule writer · 400-token cap + keyword matching · 59 test moments from unseen tasks

CHANGED: **which past chats are saved**

Every condition picked from the same 100 past chats, then faced 59 exact moments where earlier AI agents went wrong, on tasks it had never seen. A repeat means the agent got that same moment wrong too. Lower is better.



How to read the p-values: compared case by case with no memory, our memory avoided 9 mistakes that the no-memory agent made, and made 2 that it avoided (exact McNemar test, p = 0.065). A p-value below 0.05 is the usual bar for "very unlikely to be luck."

06 · FINDING THE MOOD SHIFT

Which detector spots "frustrated → satisfied"?

600 real human chats labeled by people (150 per box, 2,709 replies). F1 for the "frustrated → satisfied" box, plus speed per reply.

Laya runs free on a laptop CPU. The agent test used Flash-Lite instead, because agent chats also contain tool errors and store rules that it can read.

07 · LEARNING OVER TIME

More past chats, fewer repeats?

Repeat rate after the agent has seen 0, 25, 50, or 100 past chats.

With only 59 tests, a swing of 2 to 3 cases moves a line by 3 to 5 points. Saving every chat reached 32% at 25 chats, then rose to 46% and 41% as more chats piled in. Fixed-only memory helped once it held about 18 rules (100 chats).

08 · ALL THE NUMBERS

Full results (59 test moments, 31 correct-action checks)

CONDITION	REPEAT RATE	REPEATED	AIRLINE	RETAIL	CORRECT ACTIONS KEPT	RULES SAVED	TOKENS USED	P VS OURS
No memory	44.1%	26 / 59	50.0%	40.0%	27 / 31	0	0	0.065
Save every chat	40.7%	24 / 59	50.0%	34.3%	28 / 31	50	335	0.332
Smooth chats only	40.7%	24 / 59	45.8%	37.1%	26 / 31	16	259	0.227
Unresolved chats only	37.3%	22 / 59	50.0%	28.6%	27 / 31	20	232	0.581
Fixed chats only (ours)	**32.2%**	**19 / 59**	**33.3%**	**31.4%**	**27 / 31**	**18**	**280**	

Rules saved = average rules in MEMORY.md per fold after 100 chats. Tokens used = average memory text added to the agent's instructions. p = exact McNemar test on the same 59 paired cases.

09 · INSIDE MEMORY.MD (AIRLINE, FIRST RULES IN EACH FILE)

Same rule writer, different lessons

Learned from unresolved chats — 36 rules saved

- When a customer requests an escalation or transfer to a human agent: `AVOID` Ignoring or failing to immediately process the escalation request; `INSTEAD` Acknowledge the request promptly and initiate the transfer to a human representative.
- When a customer complains about a flight delay in a recent reservation without providing flight details: `AVOID` Asking the customer to identify the specific reservation or flight before checking your system; `INSTEAD` Inspect their recent reservations and flight statuses directly to locate the delayed flight yourself.
- When a customer requests compensation for a canceled flight that cannot be verified in system records: `AVOID` Repeatedly debating account records or refusing help without offering escalation; `INSTEAD` Clearly explain verification limits and offer escalation or transfer to human support promptly.

Learned from fixed chats (ours) — 13 rules saved

- When A customer requests multiple changes including transferring a ticket or changing the passenger name to another person: `AVOID` Searching flights, seats, or bag fees before addressing whether ticket transfer is permitted; `INSTEAD` Immediately inform the customer that tickets cannot be transferred to another person before checking other changes.
- When a customer requests to remove or change the number of passengers on an existing reservation: `AVOID` Attempting to modify passenger count directly or offering partial passenger cancellation; `INSTEAD` Inform the customer that passenger count cannot be changed and offer to cancel and rebook if permitted.
- When Customer claims more passengers exist on a reservation than shown in records: `AVOID` Immediately issuing compensation for extra passengers or arguing without re-verifying; `INSTEAD` Re-check reservation details and politely clarify the exact passenger list documented in the system.

Rules that tell the agent to hand the customer to a human: 3 of the first 7 unresolved rules, 0 of the first 6 fixed rules. The store rules allow a hand-off only when the agent truly cannot help.

10 · ONE MOMENT IT NOW GETS RIGHT (AIRLINE)

Before and after memory

CUSTOMER	"I really want to make this work, but I can't go over $1000 for the extra cost, and I need both the date change and the business class upgrade for all passengers. Is…"
NO MEMORY	calls `transfer_to_human_agents`
RULE PULLED FROM MEMORY.MD	`When Offering cabin upgrade options for a multi-segment reservation: AVOID Offering single-segment cabin upgrades when policy requires all segments to share the same cabin class; INSTEAD Present only upgrade options that apply the chosen cabin class across all reservation segments.`
WITH OUR MEMORY	"Our flight prices and fare differences are strictly system-determined based on current availability and cabin class rates, and we do not have unpublished fares or…"

11 · TAKEAWAYS

What we learned

1. **Mood is a free teacher.** When a frustrated customer ends up satisfied, the chat holds both the mistake and the fix, even when no error message was logged.
2. **Choose what to remember.** Same rule writer, same 400 tokens: fixed chats only had the fewest repeats (19 of 59), versus 22 for unresolved chats, 24 for every chat, and 26 with no memory.
3. **Small and simple works.** A plain text file, keyword matching, and a free CPU mood model: 1,218 lines of Python.

Limits. 59 test moments is a small sample: the gain over no memory is close to the usual bar (p = 0.065), and the gaps between memory types are not statistically significant yet. The customers in tau2-bench are simulated, and we tested one agent model. **Next:** more test tasks, real customers, and the free Laya detector inside the agent test.

Data: microsoft/WildFeedback (ODC-By 1.0), sierra-research/tau2-bench (MIT) · Models: Gemini 3.8 Flash, Gemini 3.5 Flash-Lite, Laya, GoEmotions, VADER

All numbers read from results/esm_results.json · 1,218 lines of experiment code, 123 lines of tests